

Data Centre Design Checklist

Pakistan-Specific Engineering — What works in Lahore, Karachi, and Islamabad

Data Centre Design Checklist: Pakistan-Specific Engineering

Not a global template—this is what actually works in Lahore, Karachi, and Islamabad

■ Core Design Philosophy for Pakistan

Global Best Practice	Pakistan Reality	Design Response
"Design for 35°C ambient"	45°C+ sustained (Lahore summer peaks)	Cooling capacity derated 40% at 45°C → oversize by 60%
"Annual maintenance"	Monsoon degradation in weeks (June–Sept)	Quarterly validation protocols built into design
"Standard AC units"	Fail at >70% RH (monsoons = 80–95% RH)	Precision CRAC/CRAH mandatory; no residential-grade units
"Dust filters changed annually"	Clog in 72 hours (Lahore industrial zones)	IP54 cabinets + monthly filter validation + particulate sensors
"24hr emergency response"	4-hour survival threshold during monsoons	On-site standby engineers June–Sept; portable cooling pre-staged

■ *First principle: Design for monsoon survival, not textbook efficiency. A DC that works in Singapore will fail catastrophically in Lahore.*

PHASED DESIGN CHECKLIST

■ PHASE 1: SITE SELECTION & LOCATION (Non-Negotiable)

Consideration	Pakistan-Specific Requirement	Failure Consequence
Flood Risk	≥1.5m above 100-year flood plain; no basement/server rooms below grade	Monsoon flooding → total facility loss (2022 Lahore floods destroyed 14 DCs)
Grid Stability	Dual utility feeds from different substations (not just different poles)	Single substation failure = 12hr outage (common during monsoons)
Dust Exposure	Avoid industrial zones within 2km (textile mills, brick kilns)	Filter clog rate ↑300% → thermal shutdowns
Ambient Temperature	Rooftop equipment shaded; no west-facing walls (solar loading)	Cooling capacity ↓40% at 45°C ambient
Access Roads	All-weather access during monsoons (no unpaved approaches)	Engineer cannot reach site during critical failure
■ Red Flag Sites to Reject:		
Ground-floor locations in flood-prone areas (e.g., Lahore's Township during monsoons)		

Single utility feed from overloaded substation (e.g., Karachi's SITE area)		
Rooftop exposure to western sun without shading		

■ **PHASE 2: POWER ARCHITECTURE (Load-Shedding Hardened)**

Component

Utility Feeds

UPS Sizing

Battery Type

Generator Fuel

Cooling Power

■ *Critical Integration: Power design must include cooling plant runtime in generator sizing. Most Pakistani DCs size generators for IT load only → cooling fails during extended outages.*

■ **PHASE 3: COOLING DESIGN (Monsoon-Hardened)**

Component

Cooling Type

Condensate Management

Airflow Containment

Underfloor Plenum

Redundancy

Emergency Cooling

■■ Monsoon Protocol: All cooling designs must pass thermal failure simulation at 45°C ambient + 90% RH before commissioning.

■ **PHASE 4: PHYSICAL INFRASTRUCTURE (Dust & Flood Hardened)**

Component

Cabinets

Raised Floor

Room Integrity

Water Detection

Cabling Pathways

■ *Dust Reality Check: Lahore industrial zones = 180–220 µg/m³ PM10 (WHO safe limit: 50 µg/m³). Standard DC designs assume "clean air" — they will fail here.*

■ **PHASE 5: MONITORING & AUTOMATION (Blind Spot Elimination)**

Requirement

Thermal Sensors

Humidity Sensors

Power Monitoring

Leak Detection

Telemetry Resilience

■ Non-Negotiable Metric: Zero blind spots. If you can't monitor it, you can't protect it.

■ **PHASE 6: REGULATORY COMPLIANCE (SBP/SECP Mandates)**

Requirement

SBP Circular DRD2-141

Room Integrity Testing

Data Sovereignty

Fire Suppression

Energy Reporting

■■ Penalty Risk: SBP fines up to PKR 50M for non-compliance + license suspension.

COST VS. RISK TRADEOFFS (SME vs. Enterprise)

Design Element

Cooling Redundancy

Generator Runtime

Room Integrity

Dust Protection

Emergency Response

■ *SME Survival Strategy: Prioritise monsoon-hardened cooling and generator runtime over redundancy. A single cooling unit that survives monsoons beats N+1 that fails during humidity spikes.*

INTEGRATION WITH SERVER CONTINUITY SUITE

Design Element

Cooling Stability

Power Quality

Dust Control

Thermal Monitoring

■ Critical Rule: Never sell ServerLife Extend™ without validating cooling/power design first. 31% of ServerLife Extend™ SLA breaches originate in facility-layer failures (cooling/power), not server hardware.

FINAL DESIGN VALIDATION CHECKLIST

Before commissioning, verify:

Monsoon Simulation Test: Run cooling at 45°C ambient + 90% RH for 72 hours

Load-Shedding Drill: Simulate 12-hour grid outage with generator-powered cooling

Dust Ingress Test: Particulate count <50 µg/m³ after 30 days in Lahore/Karachi

Room Integrity: ≥10 minutes retention at 50 Pa (NFPA 2001) — retest quarterly June–Sept

Thermal Failure Simulation: Staged cooling failure (primary → secondary → portable) with zero hotspots >27°C

SBP Compliance Audit: Documentation package ready for regulator inspection

Bottom Line

Designing a Pakistani data centre isn't about copying global templates. It's about engineering for monsoons, dust, and grid instability first — efficiency second. A DC that survives Lahore's monsoons and 45°C summers will be efficient by default. One designed for "textbook conditions" will fail catastrophically here.

Your design mantra: "If it doesn't survive monsoon season, it doesn't belong in Pakistan."